

Graph Neural Networks for Recommender Systems

Ritish Shrira0 (IMT2022003)

Mallikarjun Chakoti (IMT2022116)

Shashwat Sabharwal (IMT2022573)

April 1, 2026

The Goal

To evaluate the performance of Graph Neural Networks (GNNs) on **Recommender System** tasks using real-world datasets.

The Task: Link Prediction

- **Objective:** Predict missing edges between nodes.
- **Real-world analogy:**
 - Will User u watch Movie m ?
 - Will User u visit Business b ?
 - Is Product A related to Product B ?

Dataset Selection & Semantics

We selected the following datasets for our task:

Dataset	Graph Type	# Nodes	# Edges	Feat.	Classes
1. MovieLens 1M	Heterogeneous	9,923	1,000,209	18 / 30	Link Prediction (2)
<i>Nodes: 6,040 Users, 3,883 Movies. Task: Predict user-movie interactions.</i>					
2. Yelp	Heterogeneous	716,847	13,954,819	300	Link Prediction(2)
<i>Nodes: Users & Businesses. Task: Predict customer-business interactions.</i>					
3. Amazon (Computers)	Homogeneous	13,752	491,722	767	10
<i>Nodes: Products. Task: Product category prediction & co-purchase modeling.</i>					

Why Node Features AND Structure?

Q: Why are both strictly necessary for this task?

1. Node Features (Content)

- Provide **semantic identity**.
- *Limitation*: A user's demographic alone cannot predict exactly which specific movie they will watch next without history.

2. Graph Structure (Context)

- Captures **collaborative behavior** (e.g., "People who bought X also bought Y").
- *Limitation*: Cannot handle "Cold Start" (new items with no edges) without features.

Conclusion: Recommendation requires both *personal attributes* AND *relational context*.

Q: How do node features complement the graph structure?

① Enriching Message Passing:

- Structure defines *where* information flows (the neighborhood).
- Features define *what* information flows (the semantic payload).

② Multi-hop Reasoning:

- A GNN aggregates neighbor features to generate an embedding.
- *Example:* If a user is connected to many "Horror" movies (feature), the user node learns a "Horror-fan" representation (structure + feature), even if the user has no explicit "Horror" tag.

Proposed Methodology

We will perform an Empirical Comparison on Link Prediction using:

1. GCN (Graph Convolutional Network) Baseline approach.
2. GraphSAGE (Sample and Aggregate) Good for large datasets where full-batch training is expensive.
3. GAT (Graph Attention Network) Uses attention mechanisms to weigh neighbors differently. Crucial for recommendation.